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Small Scale Localized Maintenance of Industrial Infrastructure using Autonomous UAVs

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Abstract— **The use of autonomous unmanned aerial vehicles (UAVs) for small scale maintenance tasks such as blasting, painting and ultrasonic thickness (UT) measurement of industrial infrastructure surfaces is considered. A number of modest but new improvements such as sensing using a 3D non-linear Lidar, unique navigation implementation, filtering using a predictive sliding window filter with mutating window size, coverage planning using custom parametric function and pseudo-velocity control are made which lead to an appreciable increase in performance of the integrated system. This allows the UAV to achieve higher levels of operating capability than currently reported. These improvements are reported in detail. Real world experiments are conducted which demonstrate the feasibility of endowing autonomy to UAVs for successfully executing small maintenance tasks when coating thickness requirements are in the order of hundreds of microns. Social impacts include improving worker safety conditions when performing inspection and maintenance at heights by eliminating scaffolding construction and reducing manual effort.**

I. INTRODUCTION

The use of autonomous aerial vehicles for the automated and rapid deployment of small-scale or specialized missions involved in the maintenance of infrastructure in the Oil and Gas industry is considered. Two specific application areas that feature very prominently in the industry are 1) Ultrasonic Testing (UT) and 2) Blasting and Painting and these are the foci of attention in this paper.

In UT applications, thickness measurement of various steel-based infrastructure such as storage tanks, pipelines etc are a routine task since predictive maintenance and prevention of rupture or leakage are of utmost importance. Here, an ultrasonic wave [1] is emitted the amplitude of the returned echo indicates the thickness of the surface which can be calculated in the order of millimeters [2], [3]. Currently, UT measurements are performed manually using handheld devices for low-rise infrastructure. For high-rise infrastructure, the UT measurement device is either installed on an aerial vehicle that is then manually operated by an expert pilot [4] or the handheld device is then carried by maintenance personnel who climb specially-constructed scaffolding to reach the required measurement heights. They key objectives for the autonomous UT aerial vehicle would be to approach these spots, establish contact using the UT boom and hover for approximately 3-5 seconds to acquire a reading.

Blasting and Painting are used for spot-maintenance of small, bounded areas of infrastructure that require fresh coats of paint to ensure longevity. Certain spots corrode faster due to any number of reasons such as material inconsistency, uneven distribution of paint coats in prior maintenance etc. These spots are normally bounded in size, thus better solutions than erecting an entire scaffolding for the treatment of a single spot is needed. The blasting phase is responsible for surface preparation using abrasive garnet. This is analogous to polishing or sanding a surface prior to painting. The subsequent painting phase is then responsible for the actual application of paint coating. The key objective is to approach a specific location, on for example a high-rise storage tank, maintain a constant distance from the target surface and constant smooth sideways velocity mission for the duration of the painting. The small, localized area that requires fresh paint application has been identified previously through other means. Thus, the location and area size has been determined beforehand and is supplied to the aerial vehicle at launch time as mission parameters.

It is therefore desirable to develop autonomous aerial vehicles that can perform these tasks autonomously with high reliability and robustness. Such automation directly reduces maintenance time and costs since the current procedures involves construction of scaffolding around the target and personnel charges. An aerial vehicle such as a multicoper provides the ideal platform to perform spot UT measurements and spot painting on high-rise infrastructure. In the architecture proposed, a single aerial platform is used for both applications with interchangeable and application-specific booms that are installed for a specific mission. The software system is common to both applications since they share much of the same navigation (global and local) behavior. The only difference in navigation behavior between the two applications is that the UT aerial vehicle makes contact while Blasting and Painting aerial vehicle approaches the target surface but does not make contact. It then performs simple coverage planning of an area whose location and size has been determined and specified at launch time. We propose fully autonomous systems instead of manual flying as in [5].

Although UAVs and associated technologies can be said to be matured fields, the missions presented in this paper present additional technical challenges. Outdoor operating environments of industrial plants are characterized by strong sunlight, unpredictable wind conditions caused by gusts and downdraught around tall structures and high safety guarantees. The margin of allowable error is also smaller compared to non-industrial applications. The work described here therefore includes new improvements related to filtering/denoising using an improved filter, coverage motion trajec-

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tory generation using custom parametric function, 3D point cloud acquisition using a MEMs 3D Lidar with nonlinear scanning pattern, improved implementation of navigation techniques and improvements in system integration that to our knowledge are new. Work done in indoor environments such as in [6] are not salable or transferable to real-world industrial environments for reasons that will be explained in the relevant sections in the paper.

II. UAV Overview

A. Platform

The DJI M600 is used as the base of the aerial platform. The octocopter configuration is favored to a quadcopter configuration since additional rotors provide increased stability and redundancy in the event a rotor hits part of the environment. Obviously, it also provides increased thrust. The M600 weighs 4kg and has a takeoff weight of 9.1kg and maximum weight of 10kg with a flight time of 30 minutes. Kinematic and dynamical models are widely-available in literature e.g [7], [8], [9] so they are not duplicated here.

B. 3D MEMs Lidar

The selected Lidar - the Infinisoleil FX10 provides a high scan density thanks to its non-linear scanning pattern in contrast to most other Lidars such as the Velodyne that has a discrete number of sensing planes. It performs robustly in strong sunlight due to its tolerance level of ambient light of up to 200,000 lumens and has a small form factor. It consumes very little power since it is non-motorised, MEMs based and therefore does not contain actuators that require energizing. A full characterization study can be found in [10] that also shows the unique scanning pattern. This sensor can be considered to be a key sensing technology of this aerial vehicle given that it gives high density, noiseless and high fidelity scan data in outdoor environments under extremely strong sunlight with brightness reaching 200,000 lumens. For brevity, we define noise as spurious or ghost data points returned in the absence of any solid objects that reflect laser beams. Sensor *jitter* still exists but this presents less of a critical problem compared to noise since other Time of Flight (ToF) or depth cameras or 3D Lidars either exhibit a very low signal-to-noise (SNR) ratio in outdoor sunny environments (consumer grade ToF and/or depth sensors such as Intel Sense, Kinect or Xtion give very poor performance in outdoor environments under sunlight) or contain discrete sensing planes resulting in sparse and coarse point clouds (e.g the Velodyne). We refer the reader to [10] for detailed proofs of these statements.

The FX10 Lidar has a wide-input operating range of 12V-24V so runs of for very long periods of time with an appropriately-chosen battery. Actual examples of point clouds acquired using this Lidar are given in Section IV.

C. UT Boom

The UT inspection boom contains a Multigauge 6000 [11] drone thickness gauge from TriTex as the UT probe, an ultrasonic couplant dispenser system and a mechanical

Fig. 1: UT boom

damping and stabilization support system. The UT probe uses multiple echo technology to transmit ultrasound pulses that travel through both the coating and the underlying metal. It then analyse the returned echoes and calculate the thickness of the coating and metal separately. Given that ultrasonic waves are incapable of traveling through the air medium, it is imperative that there is total and vacuum-less contact between the UT sensor and the target surface in order to successfully acquire a thickness reading. To eliminate any micro air gaps, liquid gel is automatically dispensed from the couplant dispensor that fills in any micro air pockets the instant the UT boom makes contact with a surface. The couplant dispenser is a motorized system with a couplant reservoir containing the gel. The mechanical support system consists of an articulating joint to absorb and dampen impacts or vibration when the UAV is in contact with the target surface and to limit the pitching of the UAV by providing a large surface contact so that it does not collide into the target surface. With the UT boom, the aerial vehicle is a completely self-operated and standalone i.e untethered robot system with no restrictions in flight deployment apart from battery time limitation.

D. Blasting and Painting Boom

The painting system consists of ground painting equipment and an aerial painting boom attached to the UAV. The aerial painting boom is tethered to the ground equipment via a painting hose. The ground painting equipment consists of a pump or air compressor, a paint reservoir and a painting hose. The aerial painting boom contains a long painting nozzle that extends 1.5m away from the drone, a protective shield that protects the drone from back-scattered paint. and the spray control gun. The spray control gun controls the painting action and is either actuated manually or using a switch that is autonomously controlled by the drone on-board computer. This implies that, at least for the current experimental platform, the deployment distance is restricted by the length of the tether to the compressor.

III. Navigation

A. Global Planner

We argue that SLAM-based navigation is not repeatable or deterministic since the result depends on the existence of features in the environment and the ability of the SLAM algorithm to correctly associate detected features. There are almost no features, physical, visual or otherwise when the

aerial vehicle is at heights of 50m approaching a wall or cylindrical tank located on featureless terrain. Thus, GPS is used for coarse localization. When a forward-facing Lidar senses the target surface, precise positioning relative to the surface is then possible. The benefit of this strategy is that in-map localization of the drone does not have to be performed as it is done in SLAM. The instantaneous position of the aerial vehicle is always available via GPS readings that are streamed at 50 Hz. The destination point is specified by means of a user input click. The command velocities are generated in GPS callbacks so in the event the GPS signal is lost or not available, no command velocities are generated and the aerial vehicle immediately halts and hovers instead at its last GPS-verified position. This introduces a layer of safety against spurious motion. It is not an uncommon occurrence for wheeled mobile robots relying only on on-board SLAM to make erroneous inferences about its position due to incorrect localization results and therefore collide into walls or obstacles (which are conveniently omitted from published papers or demonstration videos). This is avoided in our system.

B. Local Planner

In the current version, the aerial vehicle halts smoothly whenever an obstacle appears in the sensing region. Active 3D avoidance without the use of external sensors such as a motion capture system are not straightforward to implement. Complete 3D avoidance will require a very large, almost the entire space continuum of sensing zone of the Lidar since for true 3D obstacle avoidance capability, an air-borne UAV has an almost infinite search space for feasible paths.

C. Global and Local Planner Implementation

The global and local planners are implemented as a subsumption architecture [12] that was first proposed in the late 1980s. A common weakness of the global and local planners of the ROS Navigation Stack, which is now more or less of an navigation research standard despite the persistence of this weakness, is that there is consistent conflict between the global and local planners. For a wheeled mobile robot, this conflict in velocity commands leads to oscillatory behavior. This motion undulation is highly undesirable for aerial applications. The solution adapted leads to clear and consistent behavior that is always robust and repeatable. Strictly speaking, there is a single integrated planner that acts as either a global and local planner since a single velocity command is ever generated.

IV. Target Surface Recognition

Let $\mathcal{P}$ be the raw point cloud acquired from the FX10 3D Lidar that returns a nonempty set of points $\{p_{i=1}^{n}\}$ arranged according to the Lissajous pattern in 3D Euclidean space. The application of linear, non-linear and planar filters on $\mathcal{P}$ are required for the removal of the ground plane and other non-target surface areas such that the final resulting point cloud $\mathcal{P}_{target}$ is that of the target surface area only. The overall filtering procedure is required to be generic enough such that it can be applied to a wide variety of surfaces without requiring any priori information. Linear filtering, in the form of band-pass filters is first performed along the three axes separately to exclude any obvious areas of zero-interest such as trees etc. in order to reduce the point cloud size. The thresholding parameters of this stage of filtering are dynamically adjustable so that they can be configured quickly for any mission.

$$\mathbf{F}_{linear_{x|y|z}} = f(\{x|y|z\}_{llimit}, \{x|y|z\}_{ulimit}) \tag{1}$$

Planar filtering removes those scan points that form the ground plane which is visible when the UAV has just taken off and is still far from the target surface. The ground plane is parametrized by planar coefficients $< a, b, c, d >$ and the distance threshold parameter for the plane is kept as coarse as possible so that ground-plane estimation is valid even for a wide variation in the terrain i.e some intentional over-filtering is present since a "thicker" ground plane is removed.

$$\mathbf{F}_{planar} = f(a, b, c, d, d_{thresh}) \tag{2}$$

The extracted point cloud is that of the target surface. It is then smoothed using non-linear outlier filtering to smoothen out the effects of sensor jitter whereby some scan points exist far from their neighbours. This occurs due to the fact that a half of laser beam may be incident on the edge of a target surface while the remaining half may be incident on an object behind the target surface thereby distorting the measurement. The level of outlier rejection is controlled by radius and minimum neighbours parameters.

$$\mathbf{F}_{non\,linear} = f(r, n_{min}) \tag{3}$$

All the above filtering procedures use smart pointer implementation [16] so that all filters can be applied on the incoming cloud in sequence such that the same input raw point cloud is recursively filtered out to retain only scan points corresponding to the target surface. This attains the fastest possible processing by eliminating unnecessary cloud copies and duplication to avoid bloated processing behavior. Mathematically,

$$\mathcal{P}_{target} = \mathbf{F}_{non\,linear}(\,\mathbf{F}_{planar}(\,\mathbf{F}_{linear}(\mathcal{P})\,)\,) \tag{4}$$

Once the target surface has been extracted, any spatial information required for the UT or Painting mission can be measured e.g shortest distance to target surface, surface curvature or linearity and surface orientation with respect to 3D Lidar and UAV. A small section of the extracted target surface, that is physically co-aligned with the UAV longitudinal axis, is further sampled and the normal vector to this area is calculated. The normal vector indicates the orientation of the target surface with respect to the UAV and is used later in yaw rate control (Fig. 2(h)).

An additional intensity-based filter is implemented for use during a blasting or painting action since the blast and paint clouds emitting from the boom nozzle are sensor-perceptible and therefore detected by the FX10 Lidar as an extrusion of the target surface as in Fig. 3(a). It is removed via the

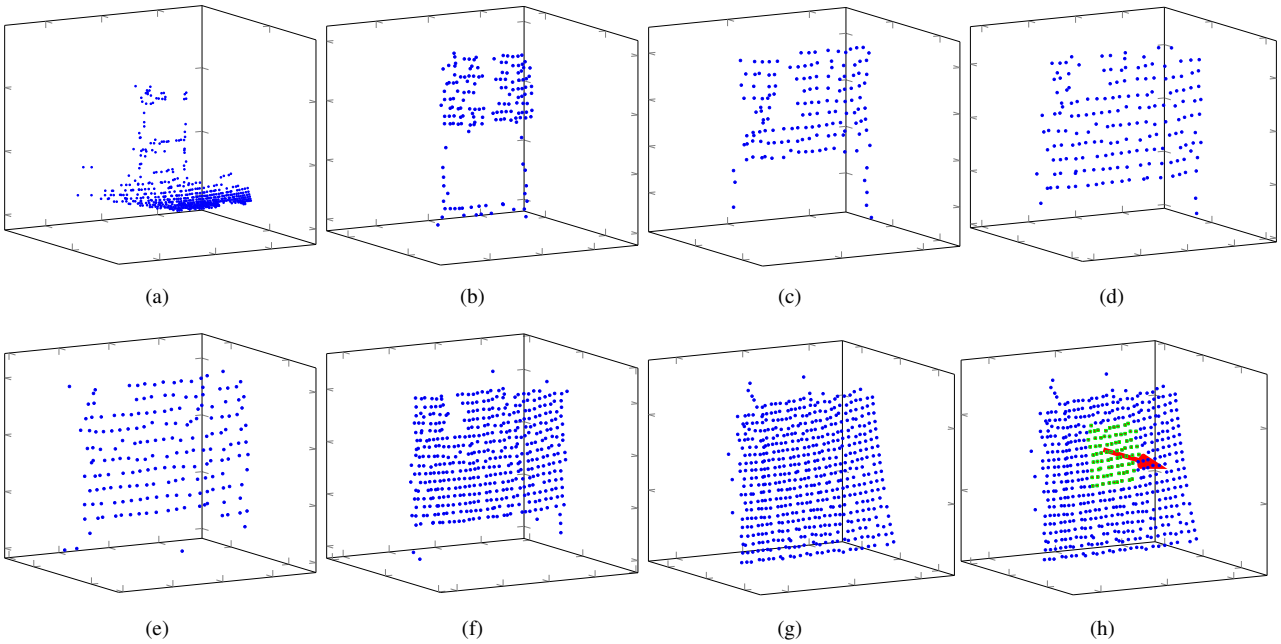

Fig. 2: Real data acquired during experiments: [a] Long range, 3D point cloud acquisition of a tower and ground plane. Due to extremely strong sunlight directly incident on and reflecting off from shiny aluminium surface, much of infrastructure surface is immeasurable from long distances. [b] Example of ground plane removal. [c]-[g] UAV acquires a target "lock". More and more of the target surface becomes visible and point cloud fidelity increases as the UAV approaches target. [h] Sampling area in green and normal vector in red

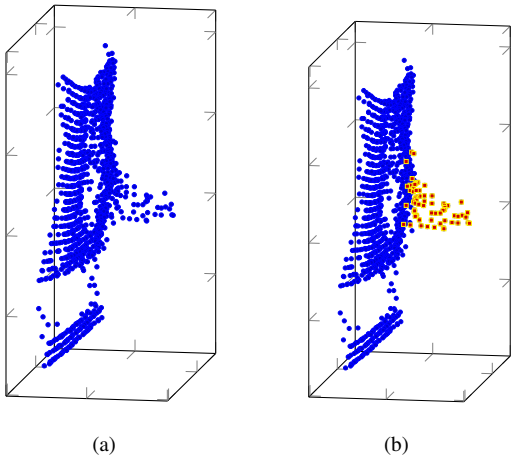

Fig. 3: Paint/blasting cloud removal via intensity field filtering

intensity filter since intensity values of the target surface and the paint or blast clouds are very distinct. This is shown in Fig. 3(b). Another option is to designate the surface as a target and track it [13] but the point cloud corresponding to the paint spray obscures the target surface to such an extent that tracking results are poor.

$$\mathbf{F}_{intensity_{x|y|z}} = f(\{x_I|y_I|z_I\}_{llimit}, \{x_I|y_I|z_I\}_{ulimit}) \quad (5)$$

The system is generic enough to work with both flat or curved surfaces since the target infrastructure ranges from building walls to storage tanks and the same procedures are used for both the UT and Painting UAV. It has been established that distance measurement can be performed with an accuracy of 2cm for an airborne UAV.

V. Control

The control strategy is derived from the mission behavior: The UAV takes off from any arbitrary position that has a direct line-of-sight path to a target surface. A destination point in the form of a coarse GPS coordinate and height is specified. The UAV takes off, attains required mission height, and flies toward the GPS coordinate that represents a staging area of sorts. From this staging area, the UAV performs precision approaching of the target surface using the Lidar. Our control strategy is to thoroughly decouple the UAV motion such that are four independent and decoupled controllers. This avoids convergence conflicts stemming from multiple controllers operating on and attempting to regulate the same control variable.

For brevity, we first explain nomenclature used by the pilot community. The *Attitude* ($\mathbf{A}$) mode refers to the mode in which the controller (either on-board flight computer or ground-based human pilot) controls raw linear velocities. In this mode, the UAV maintains stability autonomously but it is prone to drifting since the slightest wind breeze with velocity greater than instantaneous commanded velocity carries the UAV away. This mode is often described as a floating balloon. Only when the commanded velocity is much greater than wind breeze, the UAV faithfully obeys the velocity commands. In the *Position* ($\mathbf{P}$) mode, the UAV again maintains stability but in addition performs position control using GPS coordinates. The pilot commands are therefore positional or directional commands. Position mode is considered more stable (within the bounds of the GPS accuracy) since the UAV does not drift erratically at low velocities, however, positions commands are limited by GPS

resolution. Modes are switched either manually by the pilot or programmatically via a control flag in software generated by the on-board auto pilot software. Since GPS can never be guaranteed, a simple trick is used in which the control mode commences in P mode but fallbacks to A mode whenever loss of GPS is encountered.

A. Distance Control

The UAV is required to maintain a fixed distance away from a target surface and move laterally with a specific velocity in order to apply abrasive blast or an even thickness of paint. The distance controller regulates this distance parameter. An example is shown is shown with a non-linear surface. It is seen that a distance controller is valid and suffices for both linear and non-linear surfaces with the only difference yaw rate control is also required to maintain proper pose for a non-linear surface. It is straightforward to acquire point clouds from the 3D Lidar, cluster, segment and measure the shortest distance to the target surface d_{min}

A *predictive* sliding window filter with *mutable* window size is used to denoise incoming values of d_{min}. Conventional sliding window filters [14] with a fixed window size typically, for the first few incoming samples, fill in the remaining elements in the window with zero values. This creates an inaccurate average when the window has not been fully populated with samples. Thus, a mutable window length (denoted by *) is required whose width begins with 1 for the first incoming sample then elongates with each new incoming sample until it subsumes the stated window length. This way, the average is always closer to the true distance. The prediction component is next added to the filter. Given that the UAV is always in motion, any average computed even from a dynamic-sizing window is not consistent with the true average distance since the UAV has moved a small distance since the time the sliding window received the most recent sample. Thus, in our approach, we reserve a smaller window $w^*_{predict}$, inside of and as a subset of the main sliding window w^* for motion-prediction: this second window contains simulated samples of d_{min} that would occur a very short time in the future given the UAV forward motion described by current state vector $\mathbf{x}_k$. The notation x_n^{+i} is used to indicate the ith prediction of current sample x_n (= current measured d_{min}) into the future. The faster the forward UAV motion, the bigger the size of this second micro window inside the main window is with more predicted samples into a longer future given that better prediction-estimation is required since the UAV needs to halt within a finite "braking distance" without much overshoot. This results in improved estimation without the need for any tuning or choosing parameters like forgetting factors and noticeably improves the controller performance. The whole filtering pipeline described is represented by

$$\begin{aligned}
dim(w^*) &= [1:k_1] \\
dim(w^*_{predict}) &= [1:k_2] \\
k_2 &< k_1 \\
x_n^{+i} &= d_{min} - (\mathbf{x}_k \cdot i \cdot t_s) \\
w^* &= [\dots\ x_{n-2},\ x_{n-1},\ x_n,\ \underbrace{x_n^{+1},\ x_n^{+2}\ \dots}_{w^*_{predict}}] \\
d_{min\ smoothed} &= \sum w^* / dim(w^*) \\
e &= d_{ref} - d_{min\ smoothed} \\
u_{\mathbf{P}|\mathbf{A}} &= K_p\ e + K_i\ \dot{e}
\end{aligned} \tag{6}$$

Given that e subsumes signed values depending on whether the UAV has overshot the required stopping distance or not, this single control law is implemented as two separate software implementations such that two separate and distinct responses are generated depending on the direction of the UAV motion. Only one controller is ever active. The first controller provides a cautious response so that the UAV always approaches the target surface in the safest manner while the second one is characterised by an aggressive response such that the UAV backs up in the fastest manner response if it has reached (or overshot) the stopping location. The will be in the results section (Fig.10). This strategy allows the UAV to operate in windier conditions than would be possible without it since the reverse motion controller response is stronger compared to the controller response for the forward motion component. Tuning is also simplified given that the two motions are decoupled.

B. Yaw rate control

Yaw rate control is required for the drone to adapt the yaw component of its attitude to target surface curvature such that the boom remains orthogonal to the surface in the case of spherical or cylindrical surfaces. This pose is required to correctly engage the target surface both for UT readings and to apply an even coat of paint. While the distance controller maintains a fixed distance from a curved surface during lateral movement, the yaw rate controller regulates the drone yaw rate *while* maintaining a fixed distance *while* moving laterally. Simultaneous execution of the two decoupled controllers allow the drone to adapt to any curvature during lateral movement, as exemplified in Fig. 4. The UAV is consistently sampling a small frontal area directly in front of and aligned with the UT or painting boom (shown in Fig. 2(h)). It calculates the normal vector of this area with a neighbourhood radius such that the entire sampling area is covered. The normal vector orientation indicates the yaw error since in order to point orthogonally at a target surface, this normal must be parallel to and align with the UT or Painting boom. Calculated values of the normal vector are filtered by another predictive sliding mutable window filter. The controller is given by

$$\begin{aligned}
e &= \psi_{ref} - \psi_{smoothed} \\
u_{\mathbf{P}|\mathbf{A}} &= K_p\ e + K_i\ \dot{e}
\end{aligned} \tag{7}$$

This yaw rate controller is curvature agnostic. It generates the correct response even for linear surfaces due to normal-surface orthogonality, shown in Fig. 5.

C. Pseudo-Velocity Control

A (pseudo) velocity controller controls the UAV motion from an arbitrary take-off position to an approximate location of the target surface (GPS coordinate of staging area). The generated velocity is a linear function of the lower value of either the remaining distance or the distance returned by the FX10 Lidar which detects any obstacles. It has an upper bound and acceleration constraint specified. The final generated velocity value is *actually specified as a position-offset value to the controller in* $\mathbf{P}$ *mode*. This is necessary, since recall that the UAV inherently tends to drift in $\mathbf{A}$ mode. Therefore, we call this a pseudo-velocity since we achieve the desired velocity control of speeding up and slowing down but in $\mathbf{P}$ mode without drifting.

D. Altitude Control

Three statistically-independent measurement sources are used for altitude control. The on-board barometer measured altitude and GPS-measured altitudes are the first two measurements. For increased precision during coverage motion, a second, downward-facing FX10 lidar is used. The dense scan pattern of the FX10 implies that the average terrain surface is captured more accurately compared to Lidars such as the Velodyne that has n discrete sensing planes and therefore measures terrains depressions and extrusions only if they intersect with the sensing plane. An example of the dense terrain scan can be seen in Fig.2(a). The three measurements are combined using the concept of a complementary filter except that now there are three weights instead of two.

$$\begin{aligned} h &= w_1 \cdot h_{baro} + w_2 \cdot h_{gps} + w_3 \cdot h_{lidar} \\ u_{\mathbf{A}} &= K_p \; e + K_i \; \dot{e} \end{aligned} \tag{8}$$

The height component of the coverage path (planned in the subsequent section) is supplied as a reference trajectory to the altitude controller.

E. 3D Disturbance Rejection

Recoil created by the initial ejection of garnet from the nozzle creates a huge disturbance. This disturbance is characterised by a non-periodic, short and strong spike in accelerometer readings that are available at 400Hz which is adequate enough to sense any disturbances in the raw sensor data. In principle, this disturbance can be rejected using feedforward disturbance rejection by injecting a feedforward term into the control law such that it results in anticipatory response instead of pure reactive control or its effect can be minimized using adaptive control [15]. However, we find that the initial ejection-induced recoil spans such an ultra short time window that, despite being sensor-perceptible, any rejection, minimization or correction attempts to prevent the disturbance from propagating and causing major and visible physical disturbances are not successful. The only pragmatic solution is to judiciously control compressor pressure and gradually increase it such that feedback control is able to cope up with it. Wind disturbances exceeding a threshold are also difficult to handle since (1) the M600 platform, by design, is rated for wind speeds of 8km/hr and (2) wind gusts are extraordinarily complex phenomena that are multidimensional and often inestimable. Thus, currently we only rely on Position Hold as a means of disturbance rejection.

VI. Coverage Planning

Coverage planning is required to cover a target surface area during blasting and painting. Two specific coverage motions are deemed necessary and sufficient to apply adequate levels of garnet and paint: left-to-right and up-down coverage motions. These two motions, illustrated in Fig.6, are executed separately and sequentially. When combined with the above-mentioned controllers, the motions achieve correct coverage for curved surfaces as well. The coverage trajectory is produced using a trajectory generator that is specified using a custom parametric function. Functions that produce similar paths exist that can be modified and adapted, such as the square wave or Rademacher function and their variants. However, a fundamental problem is that actual discrete sampling points generated by such functions are not uniformly spaced apart. Instead, there is always a "jump" from the base of a square to its top side. This discontinuity is often concealed since these functions are plotted using line plots where all points are shown connected, like so ⎍⎍⎍ but actual data points are like ⋯⋯⋯. Supplying such a discontinuous reference trajectory to a controller causes the UAV to suddenly accelerate and transition at the discontinuity points which is undesired and dangerous. Thus, we specify the coverage trajectory as a parametric curve parametrized by t such that adjacent points are equi-distant.

$$x = t \;\; , \;\; y = \begin{cases} 0, & \text{if } t \,!= n \\ h, & \text{otherwise} \end{cases} \tag{9}$$

for $t = (1, ..., \overset{x}{\dot{n}})$. Often, t indicates time or arc length but here t is truly a latent and dummy variable. The parentheses mean the values of t are cyclic and $\overset{x}{\dot{n}}$ means n is a recurring value but only x times. Thus, t goes from 1 to n, n then recurs x times and the value then goes back to 1 and repeats the same cycle. Thus, values of t are not contiguous. Instead, they are cyclic and monotonically increasing. Equation (9) yields a coverage trajectory like the first one shown in Fig. 6 where n controls the coverage width and h and recurrence value x combined control the step height in the vertical direction. The second trajectory of Fig. 6 is just a manipulation of the first one.

VII. Experiments and Results

The test rigs, the first one of which corresponds to the point clouds of Fig. 2, are shown in Fig. 8 which consist of a 4m tower with a plate measuring 1m x 1m and 2m x 1.5m. The visualization of Fig. 9 shows the consolidated world map showing the track of the UAV, as measured by GPS, converted to UTM format and then transformed

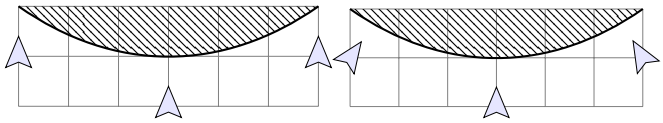

Fig. 4: [a] Example of distance control only [b] Yaw rate control combined with distance control for curved surfaces

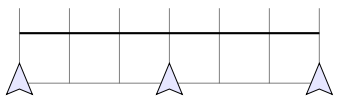

Fig. 5: Yaw rate control with linear target surface

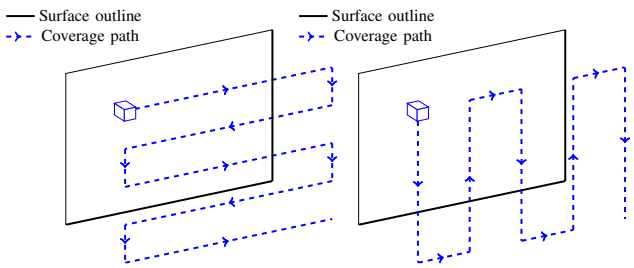

Fig. 6: Left-right and up-down coverage planning

to the world coordinate system as it approaches the target surface (partial point cloud shown) for one trial. Note that the axis ratios of the plot are unequal in order to enforce visual clarity so the length of the horizontal-like trajectory shown approaching the target surface is not proportional to altitude. GPS predictably provides reliable but coarse positioning in open spaces so it is used for coarse localization to approach the staging area without any modification to the environment since plant operators enforce restrictions that do not allow for D-RTK stations to be installed. Green points show the logged GPS positions when approaching the staging area. The drone motion was smooth and consistent even though trajectory indicates otherwise. This is attributed to the resolution/accuracy of GPS coordinates. From the staging area, the position of the UAV is measured with respect to the fixed infrastructure so cm level measurement is possible with the Lidar and cm level distance positioning and control is applied. Due to space restrictions, only results for the autonomous blasting and painting operation are shown. Results for UT inspection can be found in [4]. The plot of Fig. 10 shows the response of the distance controller as the UAV maintains (a) specified distance(s) away from the target surface. Five different setpoint distances, in reducing order, were dynamically supplied. These distances are referenced from the Lidar installation location. At the final distance of 2m (from the Lidar), the blasting/painting nozzle at the end of the boom is 30cm away from the surface. This is the required distance for applying paint. It is seen that the transition from one setpoint to another was achieved smoothly and without any overshoots or oscillation. The jagged parts in the plot correspond to the setpoint distances where the UAV halts and performs microscopic adjustments to its 3D position in order to maintain distance. It is also apparent that as the distance to the target surfaces reduces, stability increase since presumably the Lidar captures more and better quality scan data at shorter distances. There is very little overshoot. The regulation of the command velocity, when the UAV has reached a setpoint distance, is shown in Fig. 11. It clearly shows the bias of the response. The controller spends the majority of its effort in generating negative velocities to ensure that the UAV does not overshoot the stopping distance. This bias in response was a design feature. The sequence of images in Fig. 7 shows the results of an blasting operation and Fig. 12 shows two painting operations. The reduced surface thickness level after a blasting operation and increase in coating thickness after a painting operation were verified using PosiTector 6000 Probe from DeFelsko which is a coating thickness gauge. It was verified that 3-digit micron levels of coating difference could be achieved via aerial blasting and painting.

VIII. Conclusion

The use of autonomous UAVs in performing minor and localized maintenance tasks of industrial infrastructure was considered in this paper. The main contributions were providing a number of practical and robust solutions to certain challenges in sensing, navigation and control strategy. These were explained in details and backed up by real world experiments. In summary, our main finding was that while minor maintenance tasks can be executed using UAVs, it may be more challenging to satisfy technical requirements that pertain to stipulated paint thickness. Technical standards that permit a wider range of permissible thickness e.g 100-999 microns may be fulfilled easily but if the range is narrower with thinner coating e.g < 100 microns, it will be difficult to provide guarantees using aerial UAVS. Social impacts include improving worker safety conditions when performing inspection and maintenance at heights by eliminating scaffolding construction and reducing manual effort.

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References

[1] Iizuka Y. and Nagamune A. “High-resolution ultrasonic thickness measurement system for thin thickness based on frequency analysis of multiple echo”, Transactions of the Society of Instrument and Control Engineers. 1995 Feb 28; 31(2):150-6.

[2] Chauhan SK and Singh VR. “Ultrasonic Instrumentation Techniques for Thickness Measurements”. IETE Technical Review. 1992 May 1; 9(3):210-3.

[3] Ang W, Juan-hua Z, Fu-yun H, Jian-dong H and Ling W. “Measuring system for the wall thickness of pipe based on ultrasonic multisensor”, In 2009 9th International Conference on Electronic Measurement & Instruments, Aug 16, 2009, pp. 641-644

[4] V. B. S. Mahalleh, Aneesh N. Chand and Arif Rahman “Design, Implementation and Evaluation of Ultrasonic Measurement System using ROS and MQTT”, *2020 IEEE 8th Conference on Systems, Process and Control* (ICSPC 2020), Dec. 11-12, pp. 80-85, Malaysia.

[5] Rami A. Mattar and Remy Kalai “Development of a Wall-Sticking Drone for Non-Destructive Ultrasonic and Corrosion Testing”, *Drones* 2018, 2(8).

Fig. 7: Three selected sequential snapshots of an autonomous blasting experiment

Fig. 8: Test rigs

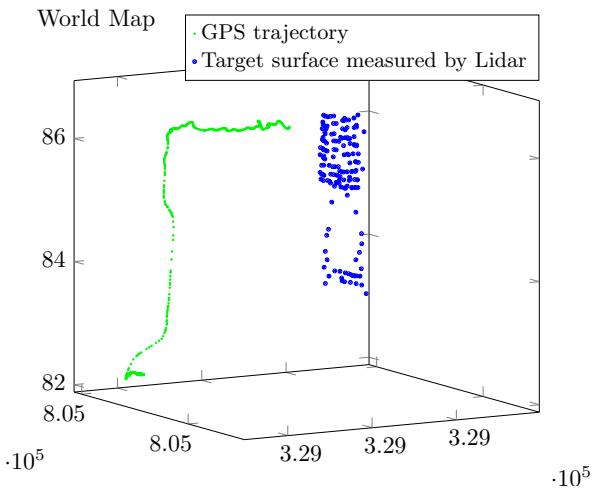

Fig. 9: World map showing drone motion towards target surface

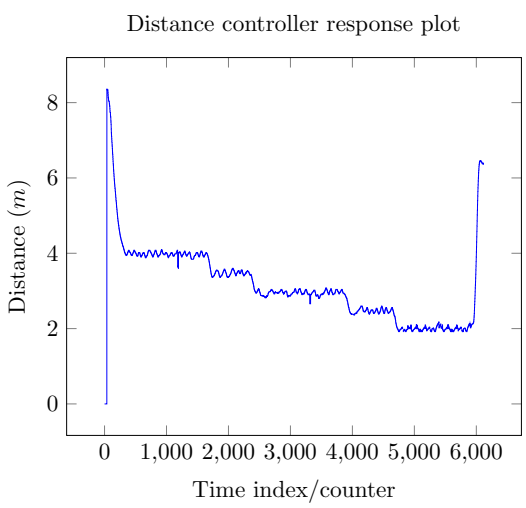

Fig. 10: Distance control for various set-points

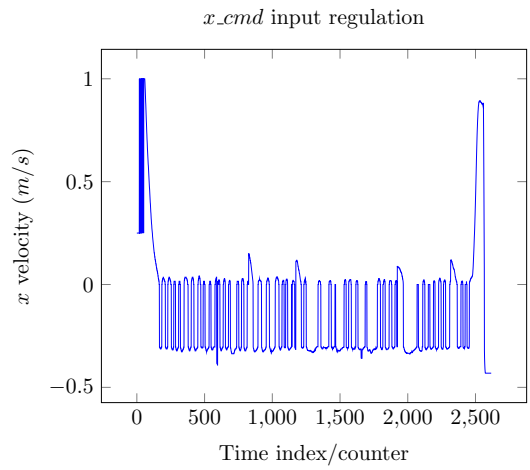

Fig. 11: Corresponding regulation of commanded velocity

Fig. 12: Painting operation

[6] A. S. Vempati et al., "PaintCopter: An Autonomous UAV for Spray Painting on 3D Surfaces", *IEEE Robotics and Automation Letters*, June 2018.

[7] D.F. de Castro, I.A.A. Prado, M. de Freitas Virgílio Pereira, D. A. dos Santos and J. M. Balthazar, "Modeling and position control of tethered octocopters" *MATEC Web Conf., Vol. 83, 2016 CSNDD 2016 - International Conference on Structural Nonlinear Dynamics and Diagnosis*

[8] He Zhu, Hong Nie, Limao Zhang, Xiaohui Wei, and Ming Zhang, "Design and assessment of octocopter drones with improved aerodynamic efficiency and performance", Aerospace Science and Technology Vol. 106, Nov. 2020

[9] Riska Analia, Susanto and Kai-Tai Song, "Fuzzy+PID attitude control of a co-axial octocopter", 2016 IEEE International Conference on Industrial Technology (ICIT), 14-17 March 2016, Taipei

[10] Aneesh N. Chand, "Performance Characterization of 3D MEMs Lidar with Nonlinear Scanning Pattern", *2021 IEEE International Conference on Intelligence and Safety for Robotics*, pp. 166-172, 4-6 March 2021, Tokoname, Japan

[11] Tritex NDT MULTIGAUGE 6000 DRONE THICKNESS GAUGE', [Online]. Available: https://www.ndt.com.au/product/tritex-ndt-multigauge-6000-drone/. [Last Accessed: 2018].

[12] Brooks, R. (1986). "A robust layered control system for a mobile robot". *IEEE Journal of Robotics and Automation*, Vol. 2, Issue 1, pp. 14–23

[13] Li L, Hu Y. "Autonomous positioning control of manipulator and fast surface fitting based on particle filter and point cloud library technology", International Journal of Advanced Robotic Systems. 2017 Oct 13; 14(5):1729881417734829.

[14] Gabe Sibley, Larry Matthies and Gaurav Sukhatme, "Sliding window filter with application to planetary landing", Journal of Field Robotics, 2010, No. 5, pp. 587-608

[15] A. N Chand, M. Kawanishi and T. Narikiyo, "Adaptive pole placement pitch angle control of a flapping-wing flying robot", Advanced Robotics, 30(16), pp. 1039-1049, 2016

[16] Munaro M, Rusu RB and Menegatti E. "3D robot perception with Point Cloud Library", Robotics and Autonomous Systems, Vol. 78, April 2016, pp. 97-99

[17] Li J, Zhou Q, Li X, Chen R and Ni K. "An improved low-noise processing methodology combined with PCL for industry inspection based on laser line scanner", Sensors, 2019 Jan; 19(15):3398.